

CS223 Computer Architecture & Organization

Control Hazards



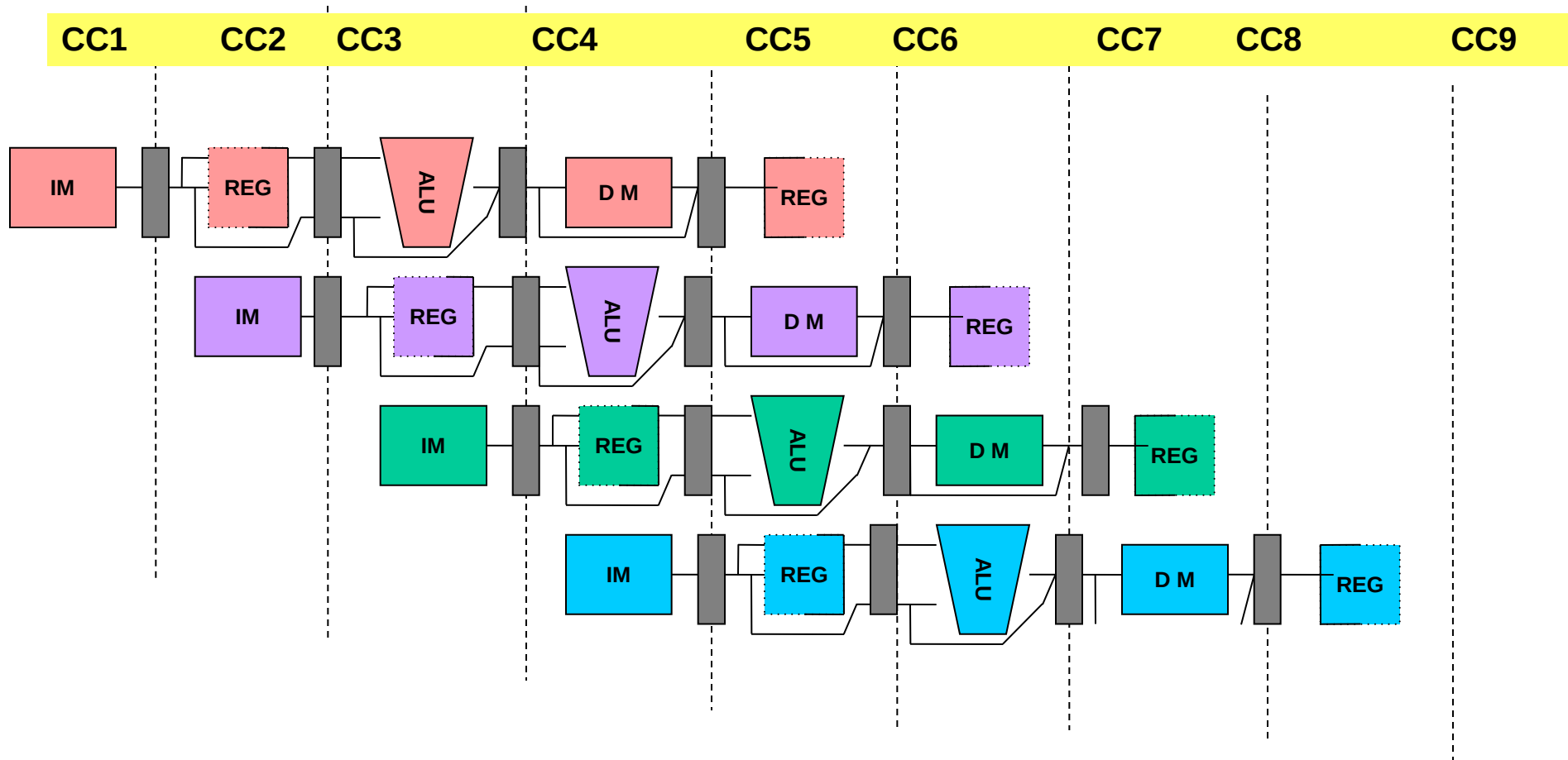
John Jose

Associate Professor

Department of Computer Science & Engineering

Indian Institute of Technology Guwahati

Visualizing Pipelining

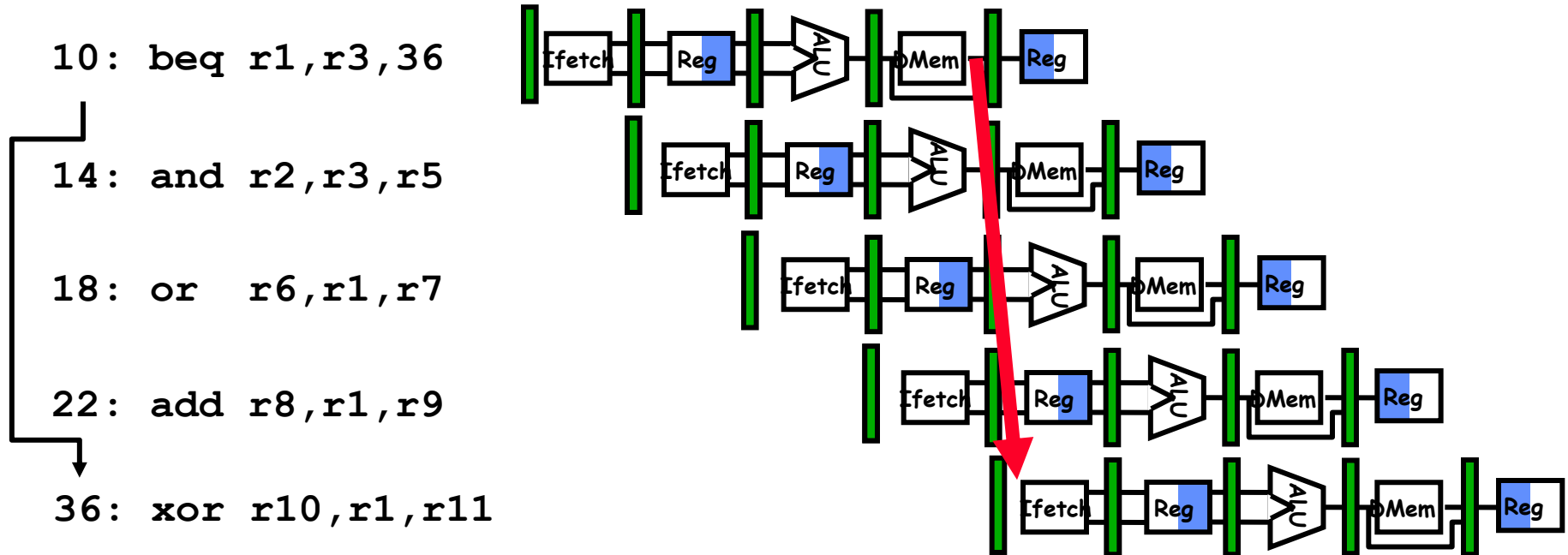


Limits to pipelining

- ❖ **Hazards:** circumstances that would cause incorrect execution if next instruction is fetched and executed
 - ❖ **Structural hazards:** Different instructions, at different stages, in the pipeline want to use the same hardware resource
 - ❖ **Data hazards:** An instruction in the pipeline requires data to be computed by a previous instruction still in the pipeline
 - ❖ **Control hazards:** Succeeding instruction, to put into pipeline, depends on the outcome of a previous branch instruction, already in pipeline

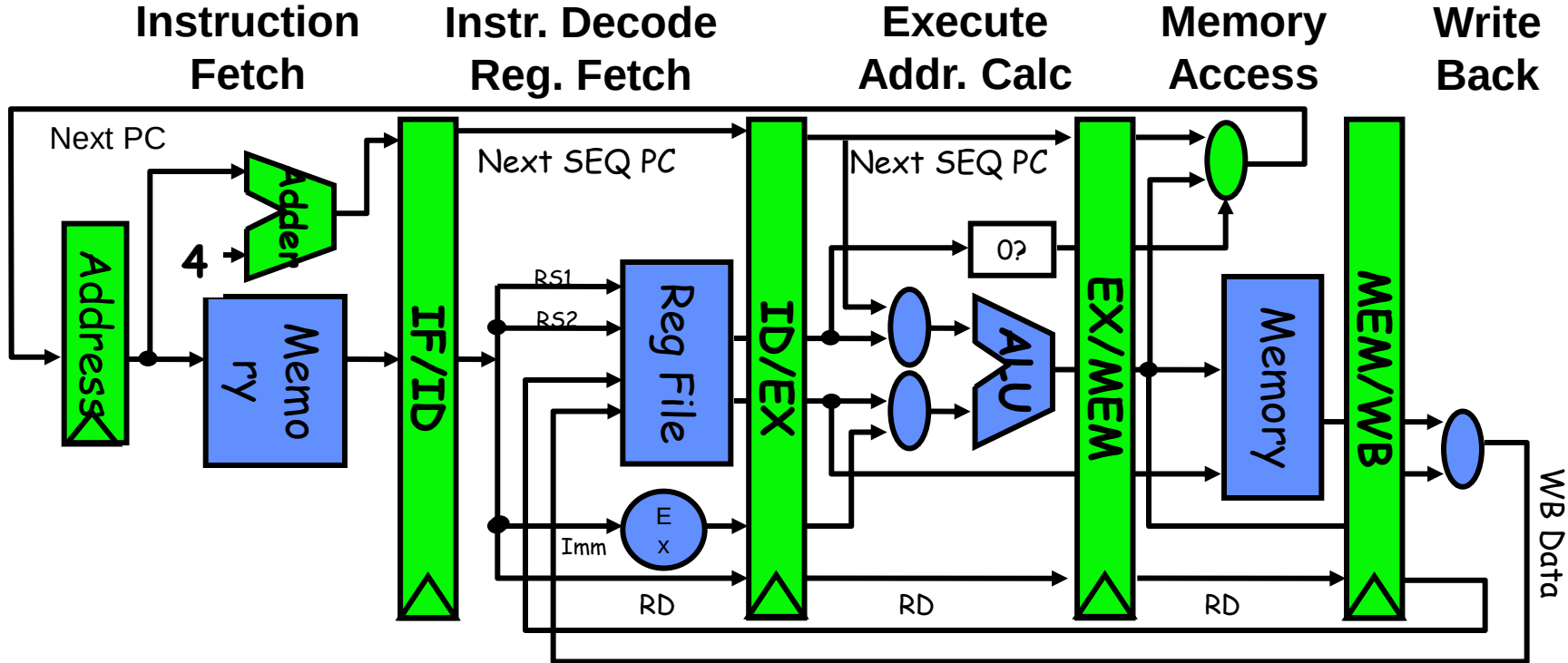
Control Hazard on Branches

=> Three Stage Stall



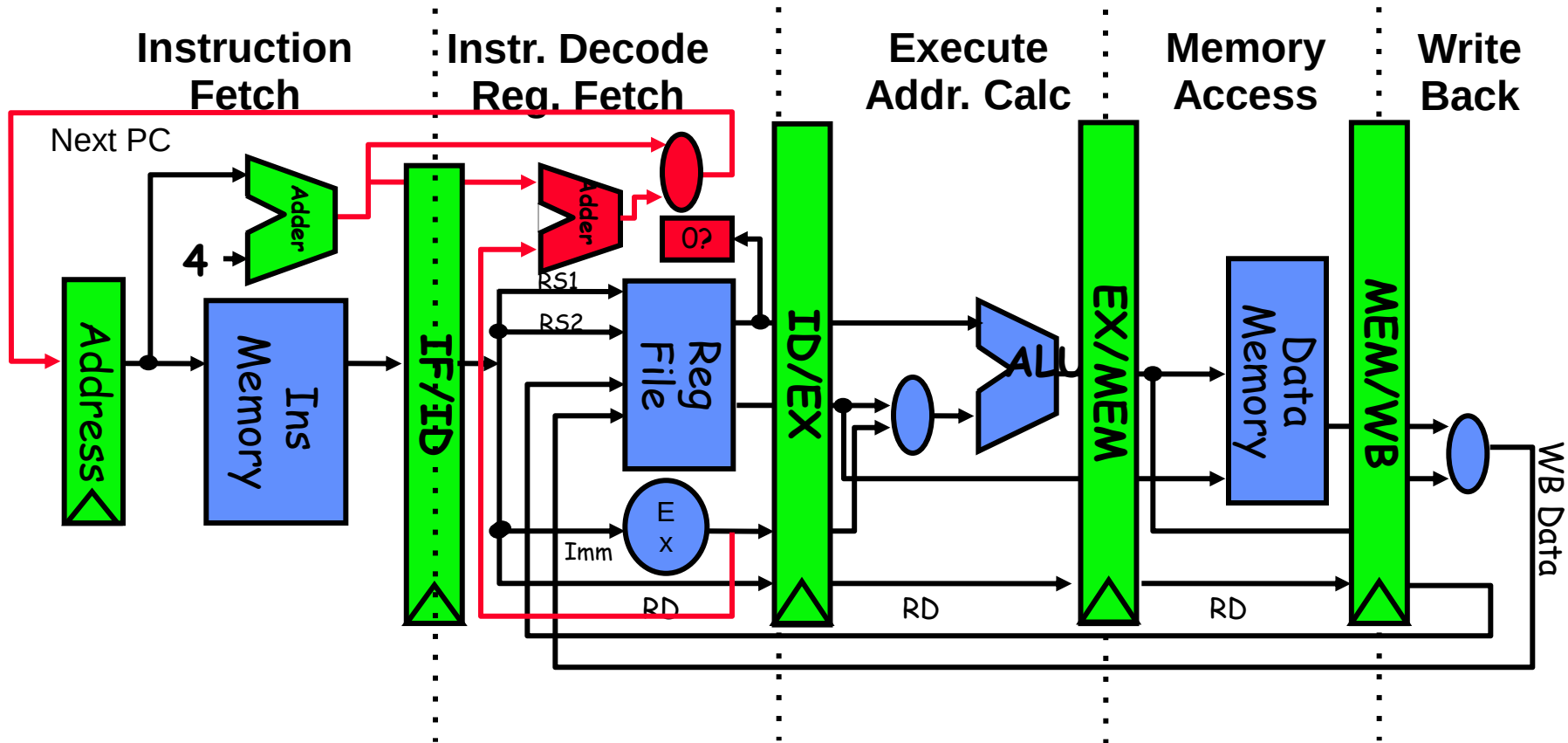
Conventional MIPS Pipeline

Branching at 4th stage



Branch Optimized MIPS Pipeline

Branching at 2nd stage



Four Branch Hazard Alternatives

#1: Stall until branch direction is clear

#2: Predict Branch Not Taken

#3: Predict Branch Taken

#4: Delayed Branch

Four Branch Hazard Alternatives

#1: Stall until branch direction is clear

#2: Predict Branch Not Taken

❖ Execute successor instructions in sequence

❖ “Squash” instructions in pipeline if branch actually taken

Untaken branch instruction	IF	ID	EX	MEM	WB				
Instruction $i + 1$		IF	ID	EX	MEM	WB			
Instruction $i + 2$			IF	ID	EX	MEM	WB		
Instruction $i + 3$				IF	ID	EX	MEM	WB	
Instruction $i + 4$					IF	ID	EX	MEM	WB
Taken branch instruction	IF	ID	EX	MEM	WB				
Instruction $i + 1$		IF	idle	idle	idle	idle			
Branch target			IF	ID	EX	MEM	WB		
Branch target + 1				IF	ID	EX	MEM	WB	
Branch target + 2					IF	ID	EX	MEM	WB

Four Branch Hazard Alternatives

#3: Predict Branch Taken

- ❖ But branch target address is not known by IF stage
- ❖ Target is known at same time as branch outcome (IDstage)
- ❖ MIPS still incurs 1 cycle branch penalty

Four Branch Hazard Alternatives

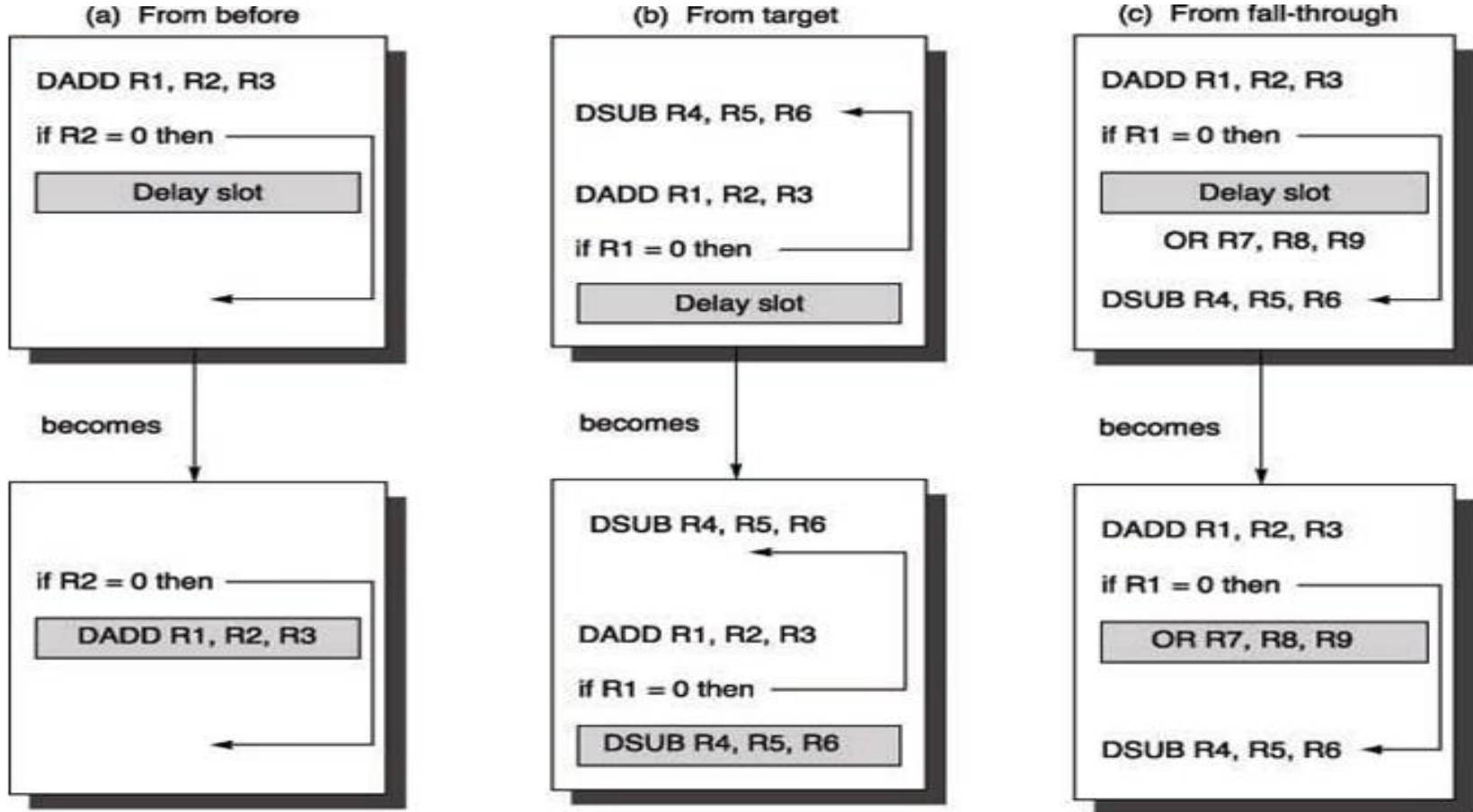
#4: Delayed Branch

- ❖ Define branch to take place **AFTER** one instruction following the branch instruction.
- ❖ 1 slot delay allows proper decision and branch target address in 5 stage pipeline (MIPS uses this approach)
- ❖ Where to get instructions to fill branch delay slot?

Untaken branch instruction	IF	ID	EX	MEM	WB				
Branch delay instruction ($i + 1$)		IF	ID	EX	MEM	WB			
Instruction $i + 2$			IF	ID	EX	MEM	WB		
Instruction $i + 3$				IF	ID	EX	MEM	WB	
Instruction $i + 4$					IF	ID	EX	MEM	WB

Taken branch instruction	IF	ID	EX	MEM	WB				
Branch delay instruction ($i + 1$)		IF	ID	EX	MEM	WB			
Branch target			IF	ID	EX	MEM	WB		
Branch target + 1				IF	ID	EX	MEM	WB	
Branch target + 2					IF	ID	EX	MEM	WB

Filling Branch Delay Slot





johnjose@iitg.ac.in
<http://www.iitg.ac.in/johnjose/>